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Occupancy behaviour and patterns: Impact on energy consumption of high-rise households in southeast China

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ABSTRACT

With the rapid urbanization in China, energy consumption and corresponding carbon emissions in residential buildings are growing. Because occupant behaviour plays a significant role in building energy performance and occupant comfort, understanding the crucial links between occupant lifestyles and energy use is the key. Most studies on the relationship between occupancy behaviour and energy usage have been conducted in public or commercial buildings. Because data is difficult to obtain and key information about existing houses is required, research on high rise residential buildings is limited. To acquire information on thermal satisfaction, residential equipment ownership and usage habits and attitude to smart meters, this paper conducted a comprehensive survey of 112 metropolitan families living in a typical booming city. A case study high-rise residential building is modelled in a building energy simulation tool. The results are compared with the actual energy bills acquired from occupants or smart meters to better understand the energy usage in this area. The results showed that a large variation in energy use could exist in different households, which is influenced by several factors such as occupancy patterns and habits, as observed in the survey. At the same time, it shows the challenge of predicting the energy use of such a building with varying internal heat gains, set points and window opening behaviours across the different households.

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1. Introduction

The rapid growth of the Chinese market, along with the significant concentration of people in cities, has resulted in a high level of urbanization and increased urban residential buildings and their associated energy consumption. China has stated that its overall CO_2 emissions will peak around 2030 and that the country will be carbon neutral by 2060. The building sector's carbon emissions are dominated by urban residential buildings, which account for more than 40% of total CO_2 emissions [1]. Between 2000 and 2018, China's urbanization rate increased by more than 20%, from 36.5% to 59.6% [2], resulting in an increase in CO_2 emissions from 289.1 million tons (Mt) to 891 Mt in the urbanized building industry [3]. Since the start of economic reform in the 1980s, China's new construction area has increased by 60 billion square meters [4], and by 2025, another 40 billion sq. m will be built [5]. Residential land makes up 25% of the total building area in cities [6], and from 2007 to 2016, the Floor Area Ratio (FAR) in residential grew, particularly in southern and central China [7], reflecting that in recent years, a large number of people from rural areas have relocated to urbanized high-rise towers. Such changes could significantly impact the residential construction sector, resulting in increased energy demand and carbon emissions [8].

Energy demand has increased dramatically for several decades, and fossil fuels still supply most of the global energy [9], while the share of renewable energy was only 11.4% in 2019 [10]. Many countries have put the development of clean fuels such as solar and wind energy on their agendas, but it still requires much effort. Recent research focuses on the integration of renewable energy into the electricity system [11], such as wind power production [12], large-scale heat pumps, and battery electric vehicles [13]. In addition, there is an increase in the deployment of smart technologies in buildings such as building automation and smart monitoring. With this trend, it is possible to reduce the consumption of fossil fuels in the building sector, accelerate renewable energy market growth, and achieve the target CO_2 emission reduction.

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Several academics have studied the impact of urban-rural variations on occupancy behaviour. According to one research, the energy demands of suburbs are larger when they get close to the city centre [14], another study found that the areas with the highest CO_2 emission were either the urban centre or rural areas [15]. Because of their understanding of ethics and duty and better consumption habits, urban residents have a larger predisposition toward energy-saving behaviours than those who live in rural areas, according to asurvey of 220 people conducted in China in 2017 [16]. Also, the income and household size link to occupants' energy use habits. Higher-income and larger households usually mean more equipment and a higher demand for living quality, which means more energy usage. On the other hand, advanced technology also encouraged improvements in energy efficiency, which called the rebound effect. According to a recent study within the household sector, the rebound impact of household energy consumption for heating is estimated to be around 20% [17].

There is no apparent link between the age of the occupants and the amount of energy consumed. Some research showed a negative link between resident age and energy use [18], while others found no significant association [19], while still, others found a positive link [20]. These disparities could be because research varied in whom they measured by age, such as any responder, household referent person, or eldest occupant. Other studies have discovered a clear connection between the age of the occupants and individual occupancy behaviours. For example, a study of the factors that influence air-conditioning operation hours in houses found a negative link between the outcomes and the occupants' age. Because older people are more likely to use natural ventilation, the air conditioner's operating hours will be reduced [21].

Switching on and off lighting, the air conditioner and heating systems, and adjusting the thermostat are all examples of occupancy behaviour [22]. According to a previous study, residents achieved energy savings ranging from 5% to 30%, with occupancy behaviour changes [23]. The previous study has shown that different occupant demands and behaviours require specific solutions, which may induce or change behaviour, and that occupant behaviour affects the flexibility and application of technology [24,25]. Numerous factors linked to occupant behaviour, including demography, social norms, and overall awareness and interest in energy saving, have been shown to influence variations [26].

Because of the large diversity of urban households, some households' power consumption can be ten times the minimum energy consuming households [27]. Changes in living conditions will cause these new urban dwellers to behave differently, resulting in a shift in energy usage. China's Research Group on Energy Use Performance investigated household energy consumption in seven cities all over the country in 2008 [28], and a similar survey was conducted in 2015 [29]. A comparison of the two studies revealed that there has been no significant change in China's building envelope system, which focuses the research objective on occupancy behaviour [30]. However, most studies of building energy use and consumption focus on public buildings like offices and commercial buildings, since the information of residents' life patterns are hard to collect due to the privacy issue and it is not easy to gather large enough samples in the same age, sex, background, and other factors.

As previously noted, China has spent the last twenty years researching the impact of urban design on building energy loads. But only a few researchers, on the other hand, have constructed comprehensive models derived from real surveys and statistics, which prevent the problem of models that are too simplistic to account for the diversity of urban forms [31]. The absence of exact knowledge of occupancy behaviour in urban residential structures leads to misunderstanding and impropriate policy and building design decisions. Unsuitable technologies have been promoted in some circumstances without a full understanding of occupant behaviour. Centralized space heating, for example, was promoted inappropriately in urban China's hot summer and cold winter zones, owing to a lack of knowledge of the substantial disparities in space heating behaviour in northern China and the hot summer and cold winter zone [32]. Furthermore, occupants are not interested or suspicious when adopting new and advanced technologies in their houses, such as intelligent measuring devices, smart meters, etc, despite the potential advantages. Hence, there should be more studies evaluating the drivers and barriers to smart technology uptake in China.

In China, the large-scale residential space-heating system is mainly applied in urban Northern China which was set by the central government in the 1950s [33]. Most research focuses on district heating when discussing Chinese residential energy use in northern China, while not much work investigated the residential energy use in southern China [34]. Despite the increasing space-heating requirement in some southern cities in recent years [35], the dominant energy consumption factor in southern China homes is the colling demands and ventilation. As incomes and standards of living increase, more people will acquire domestic devices to improve occupants' comfort and health; it will have a significant impact on the country's overall energy demand, putting pressure on electricity grids and driving up local and global emissions [36]. Therefore, it is critical to identify the key features of energy use in urban families, especially in southern China without district heating.

To get a better understanding of the key causes and promote appropriate energy efficiency methods, this study performed thorough questionnaires and a simulation of a high-rise dwelling in a typical area of urbanisation. Residents from a city in southeast China participated in the survey, and their behaviour was recorded to reflect the living routines of those who live there. The baseline is simulated with no occupants, while the case study includes the survey findings to model energy use in the specified building. The result is compared with the actual energy bills acquired from the occupants and their smart meter to better understand the energy usage in this area.

This paper aims to investigate the impact of occupant behaviour in residential high-rise buildings in southeast China to identify key factors influencing occupancy behaviour and forecast future energy consumption trends based on changing key factors, which are critical for the development of effective energy efficiency technologies and policies. This paper is structured as follows: Section 2 will provide an insight into the chosen case study and survey; the simulation software is also introduced. Section 3 showed the results and analysis of the survey and the simulated energy consumption of gained occupancy profiles; other potential measurements of smart energy management will be discussed. The conclusion will be conducted in Section 4, and the recommendations for future works will be made.

2. Methodology

2.1. The selection of case study

In developed countries like Europe, older houses typically have inadequate insulation and airtightness, performing worse than new ones. As a result, the energy load of HVAC systems in aged homes is higher than in newer homes. However, in Chinese city apartment buildings, the scenario is frequently the opposite. Even though newly constructed buildings are more energy-efficient, households often require more energy due to longer operating times and more heated/cooled areas [29].

China's percentage of recently constructed high-rise skyscrapers increased considerably after 2000, especially in the major cities [30]. Meanwhile, in 2009, the first Chinese building energy code was released as part of the country's core national policy, at a time when many buildings, particularly in large cities, had been completed [37]. As a result, many structures built around 2000 in China are considered inefficient, poor quality and without smart energy system and has the potential to be instal in the future. For this study, the population of Fuzhou, where the case study is located, was estimated to be roughly 7.7 million, with 5.4 million living in the city area [38]. For this study, a high-rise residential building constructed in 2006 was chosen as an example of buildings constructed in China's rapid development phase. Later sections will go through the specific aspects of the building envelope and operation.

For residential buildings in this area, the energy system is usually referred to the electricity system, which is supplied by power plants, and fuelled by coal, hydropower, wind power and nuclear power, according to the government yearbook in 2020 [38]. The cooling demand is supplied by the room air conditioner and usually installed in bedrooms, while heating is supplied by individual electric boilers or resistance heaters. In this case study, air conditioners are installed in bedrooms, and the kitchen and most houses don't have heaters, based on the field survey.

The chosen case study is a high-rise dwelling in Fuzhou city, one of Southeast China's major cities (Fig. 1). Fuzhou city is treated as a representative urban area in this study, whereas the surrounding countryside is treated as a representative suburb area. The structure, which was completed in 2006 and included 11 storeys, represents the typical building envelop in southern major Chinese cities during the period of urbanization. According to the survey, each apartment is roughly 100 m^2, which is the typical household size in China [39].

The case study area is classified as a humid subtropical climate zone, with hot and humid summers dominated by tropical air masses and mild winters, according to the Köppen-Geiger climate classification [41]. Space cooling is commonly utilized in the summertime, but no heating is needed in the winter in this climatic region.

In this study, relevant archival information about the chosen dwelling was obtained (Fig. 2). Architectural drawings of the building, project specifications, materials, numbers, and photos were collected and used in the subsequent analysis as well as inputs for software simulation. Building archival files were used to determine the material and thickness. Hypothetical assumptions may be utilized in some cases to apply various methodologies in the simulation, which will be the major tool for modelling energy consumption in this study. Table 1 shows the building elements as well as the U-value of the materials. As can be seen, there is no insolation layer in the building, which is usual in southern China. As explained in later sections, this could result in additional energy and time spent heating and cooling.

In China, the State Grid Corporation of China (SGCC) has built an information collection system and promoted the adoption of smart meters since 2009 [42]. Fig. 3 shows the smart meter and its onsite installation in the case study building which is on the wall of the building entrance hall. There is no in-home display, but the residents can read their data on the corresponding meter. The electricity data is collected by the smart meter from the individual electricity box of each household and further transmitted to the network. A website based interface (https://www.95598.cn/osgweb/index) is also available for customers to check energy bills, compare local energy utilization, save history electricity data, and seek potential energy-saving suggestions.

2.2. Survey of occupancy behaviour

A set of questions targeted residents of high-rise buildings were included in an online survey designed to investigate behaviours and associated occupancy patterns in the target area. The survey was first conducted in English before being translated into Chinese, with the translations overseen by the authors. It did not include the



Fig. 1. The location map of the case study in Fuzhou city, Google Map [40].



Fig. 2. The facade and layout of the case study dwelling (Provided by the author).

Table 1
The details of the case study building envelope.

Building Elements	Description	Thickness (mm)	U-value (W/m^{2}K)
External Wall	Concrete wall with no insulation	240	2.65
Internal Wall	Brick Wall with no insulation	120	2.17
Roof	Pitched roof with no insulation	240	0.98
Ground Floor	Concrete layer	400	0.70
Internal Ceiling/Floor	Timber floor on a concrete layer	150	1.42



Fig. 3. Example images of smart meter and onsite installation in the case study building.

disclosure of personal information or the discussion of sensitive topics that could have a psychological impact on the occupants. Invitations to take part in the interview are issued to residents in high-rise buildings in Fuzhou. The survey's goal is to get a broad image of Southeast China's households based on various factors: household size, ownership status, geography, and HVAC system installed. The invitation was in the form of an email, asking the resident to go to a webpage with a number and answer the survey. All survey data will be used exclusively for this project, and all data will be erased after one year.

The survey was divided into eight components. The first section provides background information on the gender and age of those who took part in the survey. The second part focuses on the dwelling's location and how people commute, as several studies mentioned in the previous section that location impacts occupancy patterns. The third section is about the size of the household, and the fourth part is about the length of time the occupants stay. The fifth part investigated how the equipment was used in respondents' homes, which was linked to internal gains. These data can be used to create an internal gain profile, which will be used for the simulation. The cooling setpoint in respondents' homes is discussed in the sixth section. The seventh part is for thermal comfort, and lastly, the window opening behaviour for ventilation. The survey provided a supplementary figure demonstrating window-opening behaviour. The time when occupants open their windows was recorded on a random workday and weekend to examine their window opening behaviour. The focus was on the bedroom, living room, and kitchen because these are the spaces where the anticipated occupants spend most of their time. Respondents can indicate an estimated timeframe for which they ventilated their rooms, or they can report the exact time if the options provided are insufficient. A survey of the occupants' attitude toward the smart meter technology is also made about the awareness of the smart meter in the building and if they are willing to change their behaviours due to the smart meter data.

A total of 262 people logged on to the website, and 112 people finished the survey questionnaire, indicating that 43% of the sample received an invitation and answered the survey. Respondents who have been willing to engage in the poll self-administered the questionnaire. Male account for 61.3% of all responses, while women account for 38.7%. The data from age groups ranging from 20 to 50 years old. Because occupants in these age groups are the most prevalent working age group and are typically the household maintainers, they will have stronger control over behaviours such as opening/closing the window, heating/colling point set, and light-switching.

2.3. The workflow of simulation

The simulation of energy consumption and other applications is presented in Fig. 4. IESVE (Virtual Environment by Integrated Environmental Solutions) was used to develop prototype models for this study. The “Model IT” module in VE is used to create models of the selected buildings, which can be examined in various ways. With the “Radiance” module for day-lighting, the “MacroFlo” module for natural ventilation, and the “Apache” thermal analysis module for both dynamic modelling of energy consumption and indoor thermal comfort [43].

The simulation's results were regarded as a realistic approximation of the real case. The software is the product of the building's systemic complexity, because of the simulation's sophisticated calculation, the accuracy of results is dependent on data inputs. The questionnaire analysis was used to configure the data in this study, which contained occupancy profiles, actives, and internal gains. The physical characteristics of fabric and façade in studied building were approximated in the building model.

Data analysis was carried out in this study, and the results were then visualized in the VistaPro tool. The first step is to collect data from literature and statistical data, which will be used to calculate and validate real energy use.

The data for calculation, comparison, and analysis comes for the second part, which comes from surveys and investigations:

- Occupancy survey on domestic energy usage in Southeast China's metropolitan homes in Fuzhou city, with the primary component of the questionnaire focusing on internal gains in the household. In IES VE, the data on occupancy behaviours will be used to develop occupancy profiles.
- Information obtained through the survey includes occupancy energy usage and internal gain profiles, and energy bills from the smart meter.
- Simulation and modelling. The software IESVE, which gives thorough insights into the case study from a micro standpoint, was used to simulate energy consumption and thermal comfort in the dwellings.

3. Results and discussion

3.1. Survey results

3.1.1. Background and general information

According to the survey of 112 households, 71% of respondents own their home, 15% rent their home, and 14% share their home with others. Most respondents' homes are between 50 and 100 square meters in size, with two or three bedrooms. In terms of the number of occupants in a household, most are three or four individuals, which matches the common Chinese family size of three occupants. The new Chinese law on having more than one kid, on the other hand, will have a considerable impact in the future [44].

3.1.2. Occupancy behaviour patterns

According to the survey, most people spend their weekday evenings and nights at home and their weekends (Fig. 5). According to the survey results, compared to people who live in cities and suburbs, more people in the city centre stay at home on weekdays and fewer on weekends. In general, on weekends, the household is all-day occupied, and on weeknights, it is occupied.

3.1.3. Internal gain profiles

According to the residential appliance/equipment survey, internal gains and energy consumption are influenced by profiles that were developed to evaluate the impact on energy consumption. The following are the ownership rate and internal gain patterns employed in the simulation (Figs. 6 and 7).

In most responders' homes, air conditioning, television, and lighting are used in the living room. On weekdays and weekends, most individuals switch on their air conditioning from 6:00 to 22:00. And they turn on television and lights in the evenings (18:00 to 8:00 next day) (Fig. 7).

Most respondents utilize air conditioning in their bedrooms while computers and lights are the main sources of internal heat gains. On both weekdays and weekends, the air conditioning is turned on, however, most people only use computers at night on weekdays. On weekdays and weekends, the light is normally turned off by the occupants during the daytime (Fig. 8).

In their kitchens, most responders have refrigerators, cookers, range hoods, and lighting. most residents leave their refrigerators on all the time. Because these home appliances function together, they are frequently used on weekday nights, and weekends all day, according to the usage schedule for the cooker and range hood. In

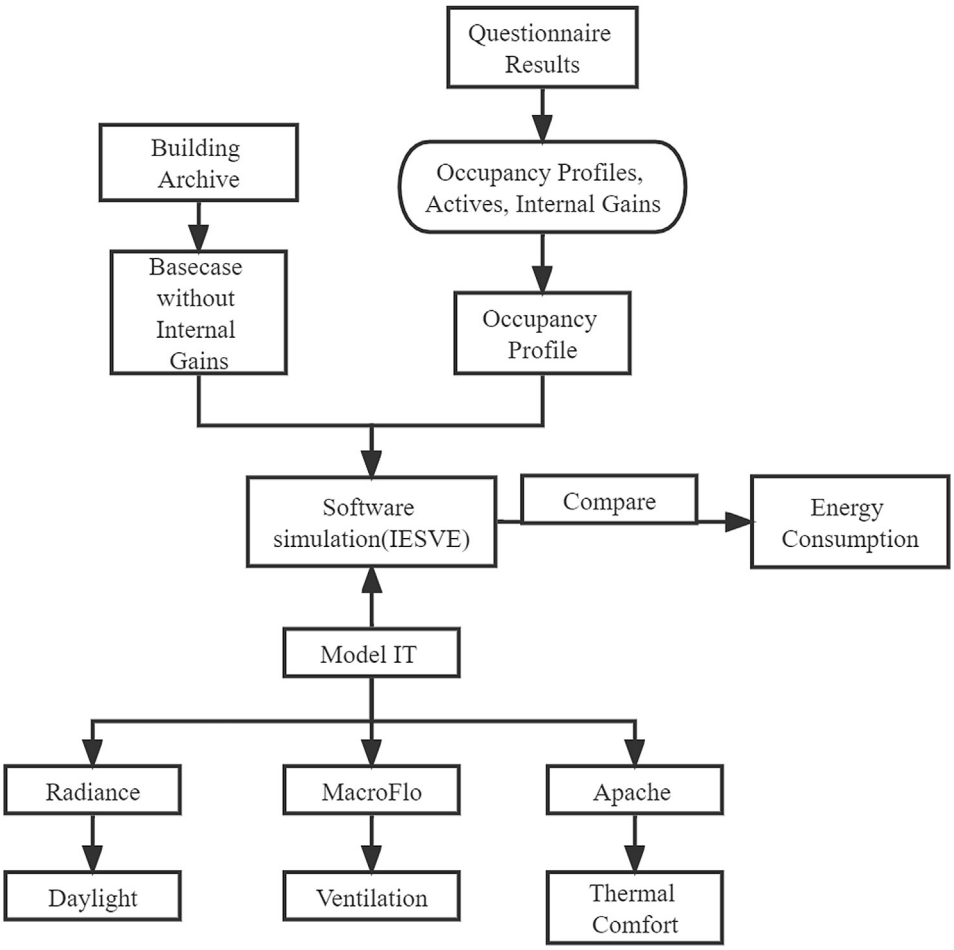


Fig. 4. The workflow of simulation in energy consumption, day-lighting and thermal comfort in this study.

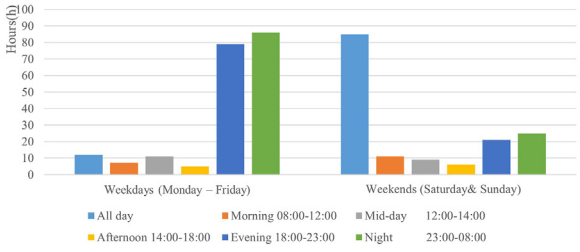


Fig. 5. The occupancy profile of stay in the house on weekdays and weekends.

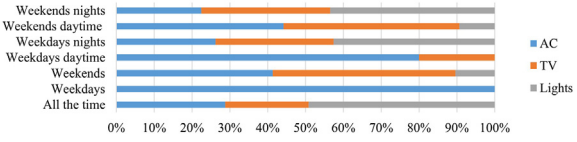


Fig. 7. The working time distribution of equipment in the living room.

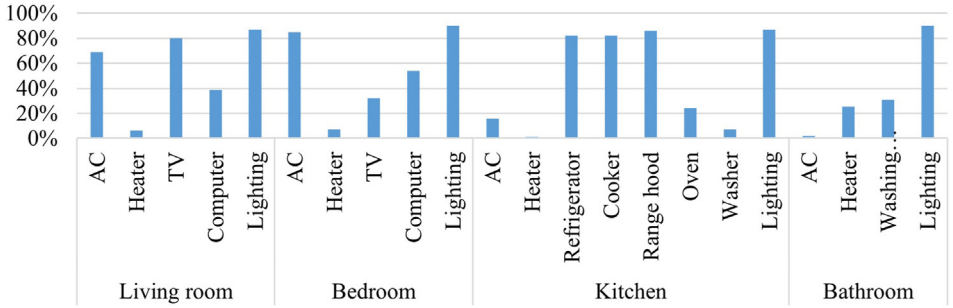


Fig. 6. The distribution of equipment, and appliances in the household according to the survey.

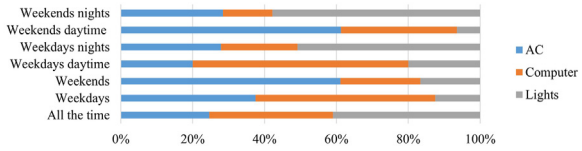


Fig. 8. The working time distribution of equipment in the bedroom.

their kitchens, occupants switch on lights at night on weekdays and weekends (Fig. 9).

On weekdays and weekends, most individuals only use lights in the restroom at night. In the survey, just about 10% of respondents use a radiator in their living room and bedroom, while about a quarter of respondents use heaters in their bathroom. This suggests that most people do not use the heater in the winter, as indicated by the climate zone's characteristics. In the summer, however, most people use air conditioners to cool their living rooms and bedrooms, with the temperature set at 30 °C. In comparison to their suburban neighbours, people set a higher cooling setpoint. In the simulation study, the cooling setpoint will be set to 30 °C.

3.1.4. Thermal satisfaction

During both summer and winter, residents were asked to rate their perceptions of thermal comfort in their homes. From “Very Dissatisfied” = 1 to “Very Satisfied” = 5, five-point rating systems were introduced. In the summertime, 46.8% of respondents reported that they are “Very Satisfied” or “Satisfied” when answering the survey. Only 6.3% said they were “Very Dissatisfied”, and another 6.3% said they were “Dissatisfied.” In the summer, the average satisfaction is 3.41, which indicates a high degree of pleasure. In wintertime, as shown in Fig. 10, the number increases: 50.4% were “Very Satisfied” or “Satisfied” with the thermal satisfaction in homes. On the other hand, 13.5% of those who lived there were dissatisfied. The average degree of thermal comfort satisfaction during the winter is 3.43, which is likewise a high level of satisfaction.

3.1.5. Ventilation and window opening behaviour

Comparisons between room types and weekdays and weekends are used to analyze window opening behaviour. The results of the interviews show that weather factors such as rain, wind, and air quality have a significant impact on patterns of window opening involvement. Opening the window can always improve the indoor air quality and provide passive cooling both physically and mentally way. It cannot operate with the air conditioner at the same time, so most people tend to open the window when they are absent. On weekdays, most people open their living room window just in the evenings (18:00–22:00), while on weekends, they open it all day. On weekdays, most responders open their bedroom windows during the day (8:00 to 22:00). Most people keep their windows open in the morning and evening on weekends. Also, most occupants prefer to keep their kitchen window open all day, both on weekdays and weekends (Fig. 11).

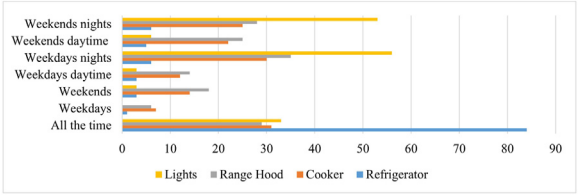


Fig. 9. The working time distribution of equipment in the kitchen.

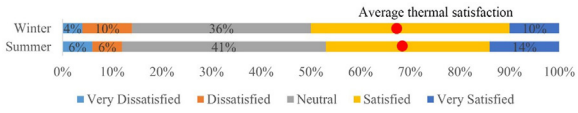


Fig. 10. Thermal comfort satisfaction in summer and winter.

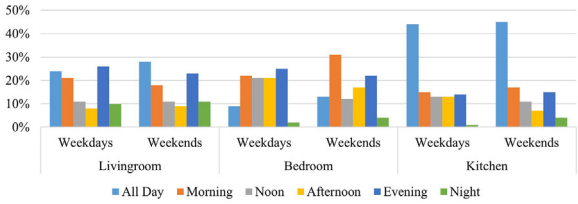


Fig. 11. Window open/close state of different rooms on weekdays and weekends.

3.1.6. Statistical correlation

The correlation between the variables above was investigated further. The investigation will aid in identifying probable links between the phenomena discussed in the previous paragraphs. The findings show a link between the occupants' age and their satisfaction with their thermal comfort during the summer. This indicates that elderly residents are more satisfied with their thermal comfort in the summer. According to a 2013 study, occupants under 30 years old are dissatisfied with their thermal comfort [45]. It is confirmed in this study by the above-mentioned survey data.

During the summer, it was also observed that there is a link between occupant age, equipment ownership, and cooling setpoint (Fig. 12, Fig. 13). It was observed that older respondents utilize less equipment, particularly PCs, and have a higher cooling setpoint in terms of thermal comfort. It also confirmed with the findings of previous studies: senior residents prefer natural ventilation to air conditioning [21].

The ownership of equipment in senior occupants' homes is different than in other age groups. For example, seniors keep their heating in their bathrooms and may not have a PC or television in their bedrooms. The usage of this equipment may have altered, influencing the profiles of internal gains. Therefore, it is difficult to model a building with every detail since it varies a lot and is a very personal part. In this study, we will select a typical occupant's profile acquired from the field survey as a case study to present the energy use behaviour in this area.

3.2. Case study simulation

This study's simulation was split into 2 parts: a base case with no heating/cooling or equipment and a case study using occupied profiles and internal gains from the prior survey. The chosen household's model workflow was previously shown in Fig. 2 to calculate the energy loads and CO_2 emissions for a high-rise dwelling. All building envelope materials, doors, windows and profiles based on survey results and structure archives were collected through the study. The outdoor air temperature and

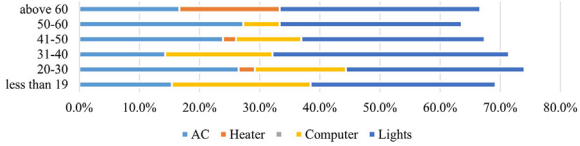


Fig. 12. The ownership of AC, heater, computer and lights in the various age group.

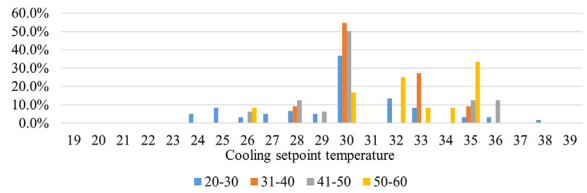


Fig. 13. The cooling setpoint temperature in different age groups' households.

climatic data were gathered using the city profile "Fuzhou." The simulated data is from January 1 to December 12 in 2016, covering the entire year to give a complete view of energy consumption.

The Ministry of Housing and Urban-Rural Development (MoHURD) classified China into five primary temperature zones for dwelling projects based on average exterior air temperatures during the summer and winter [46]. Fuzhou city is in the position of "hot summer and warm winter" zone, and according to the newest design code, JGJ 75–2012 [47], the comfortable indoor air temperature should be less than 26 °C in summer dwellings with air-conditioners, while the operative temperature ranges from 18.0 to 30.0 °C. The operating temperature is used as a comfort index when the airspeed is low. Both the ambient temperature and the mean radiant temperature are considered when determining the operating temperature.

The simulation tool, on the other hand, uses the dry resultant temperature (DRT) instead of the operative temperature. GB 50736–2012 [48], the other building design code, specifies heating/cooling settings and operating standards. Two thermal comfort situations were tested in this standard for long-occupied areas with heating/cooling. The first kind suggested air temperature from 24 °C to 26 °C, humidity ranging from 40% to 60%, and airspeeds less than 0.25 m/s. In the meanwhile, the air temperature is 26 °C–28 °C, the humidity is under 70%, and the air velocity is less than 0.3 m/s in the second category. However, there are no criteria for architectural design in these standards, which are important for HVAC system design and operation.

3.2.1. Base case simulation results

There are no HVAC, internal gains, or natural ventilation profiles assumed in the base case to explore the effects of occupancy behaviour and its effect on energy usage when compared with the case study model. Fig. 14 shows the outdoor (blue line) and indoor (red line) temperatures, it can be seen they are almost identical to each other, which is expected for a building without HVAC and internal gains. Therefore, only less than 65% of the whole year, the indoor temperature reached the comfort zone, which is 18–30 °C according to GB 50736–2012 [48]. For the other times, the HVAC system is required to provide a comfortable indoor temperature.

When focusing on the summertime indoor and outdoor temperature (see Fig. 15), the indoor air temperature is higher than the outside due to the fabric and solar heat gains. Considering the internal heat gains are not yet considered, it can be expected that the cooling demand and air condition load will be high for the building during these periods.

While the household thermal comfort appears acceptable for most of the year, with less than 65% of the time inside the comfort zone, it is not comfortable during the summer months.

3.2.2. The case study simulation results

Then the simulation results for the base case and case study models with the occupancy profiles, internal gains, equipment profiles and window-opening behaviour of typical occupants in Fuzhou city are compared.

Fig. 16 shows the comparison of the predicted indoor air temperatures in the base case and case study building models. The temperature reached the comfort zone with the HVAC system, especially in the summer. The percentage of comfort time rose to 90%, and the indoor temperature is all less than 30 °C. However, there are less than 10% uncomfortable hours in winter, which could be due to the absence of heaters in this case study, based on the survey, the occupants tolerated it.

The PMV results in Fig. 17 showed that the occupants' comfort was satisfied most of the time during the whole year, although it was a little bit cold in winter.

In particular, the comfort levels in different rooms were different (see Fig. 18), more than 90% time in bedrooms and kitchens with air conditioners is inside the comfort zone while the living room, bathrooms, and balconies in the north orientation experienced high temperatures in summer. 20% of the time, the

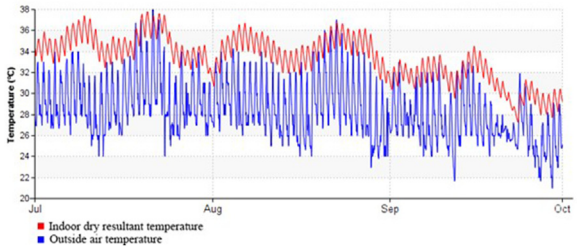


Fig. 15. The comparison of indoor and outside temperatures in summer in the base case.

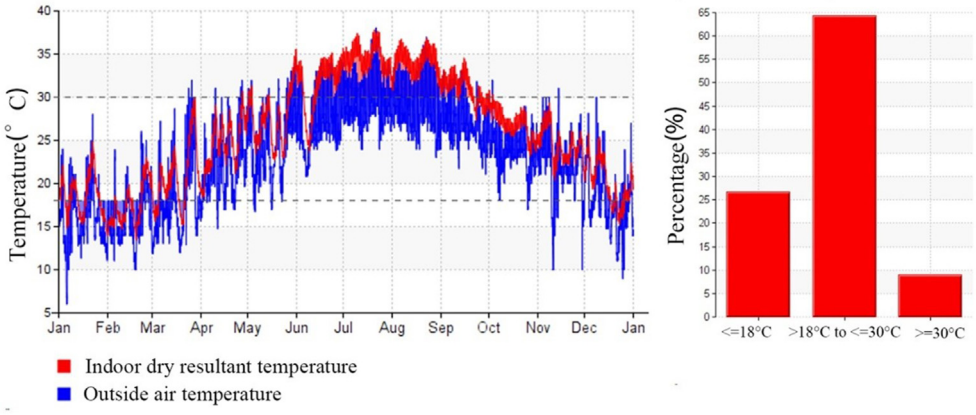


Fig. 14. Base case indoor temperature compared with the outside air temperature and the comfort time percentage for the whole year.

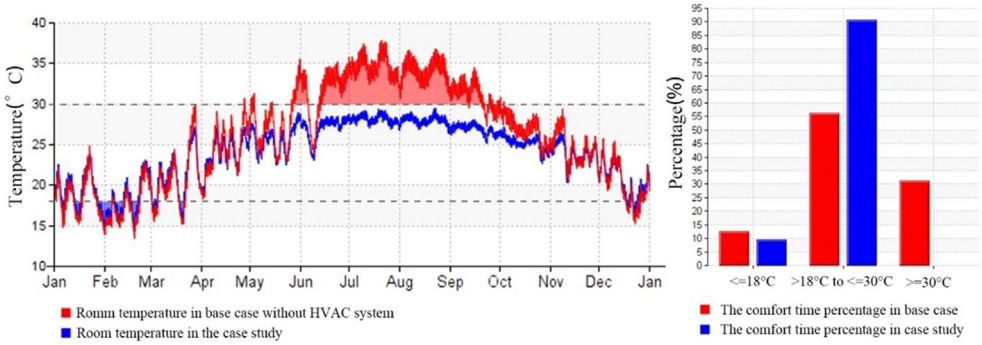


Fig. 16. Difference in the comfort temperature-time during a year between the base case and case study.

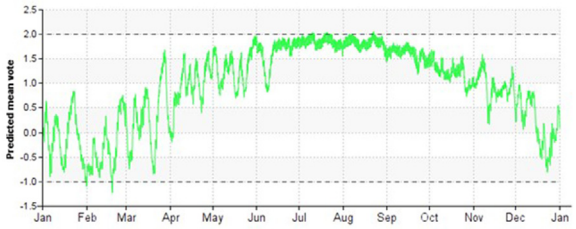


Fig. 17. The annual predicted mean vote (PMV) graph of the case study.

balcony and bathrooms are too hot in summer due to their orientation and lack of natural ventilation.

The simulation predicted that the household's total annual energy load including the HVAC system, home appliances and internal gains is 3259.80 kWh. The simulation result for the total carbon footprint for the household is 5359 kg CO_2 equivalent, with CO_2 emissions from the use of appliances/equipment accounting for 907.19 kg. A previous study investigated the energy use in China's five climate zones in 2017 and observed that the yearly average value for power use per family is 2800 kW h [49]. Our simulation result is relatively high perhaps because the case study building is in the city centre, and they tend to use more home appliances.

Two households from the case study building located on the same floor and face each other are selected and their energy meter readings for the year 2021 were compared with the simulation result (Fig. 19). The floor plan and area are identical to each other, but there are only 3 occupants in one apartment and 6 occupants in the other. Their home appliances are different accordingly.

The total energy load in household one is 2432 kWh, and the other one is 6813 kWh which are significantly different from each other, and the simulation result is closer to household one. The possible reason could be household two have more occupants and home appliances. The actual energy use is affected by several factors as in the survey before, including the number of occupants and personal energy use habits.

In this paper, the smart meter is already installed in the case study building. Some research refers the smart meter as "unprecedented insights into energy use behaviour" and has other advantages such as the avoidance of fraud and theft, better energy management, and innovative business models and pricing tariffs [50]. In our survey of the attitude toward smart meters, 68.75% of responders believed that the smart meter would help them better plan their energy usage. 62.5% of the interviewees would like to change their energy use habits, including updating the home appliances to energy-saving types and turning off lights when they leave the room. Therefore, as one component of a smart grid, a smart meter could help both environmental monitoring and governance and occupancy behaviour [51].

However, only 30% of the interviewees noticed they were using the smart meter, and most people are unfamiliar with the concept of smart grids. More than 75% of interviewees said they would read the detailed energy bills only when it changes rapidly. Also, the concern of privacy might make some people against adopting the technology [52].

The results show a typical energy use scenario in a southeast China household and provide an overall picture of thermal comfort and energy consumption in the residential house. The results also showed that a large variation in energy use could exist between two households, which is influenced by several factors such as occupancy patterns and habits as observed in the survey. At the same time, it shows the challenge of predicting the energy use of such a building with varying internal heat gains, set points and window opening behaviours across the different households. The present work shows the advantages and importance of integrating

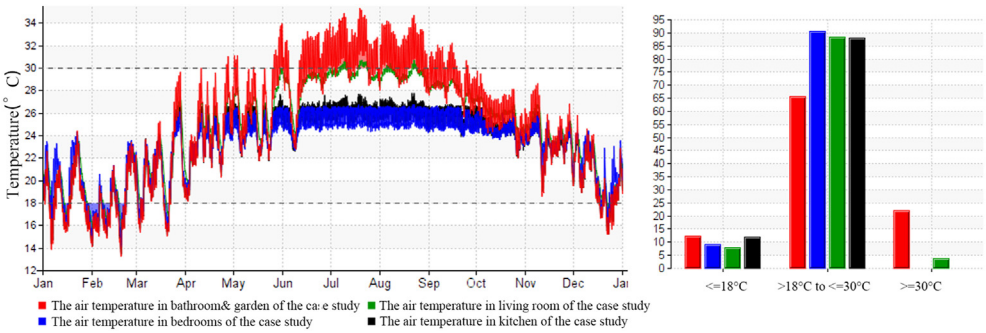
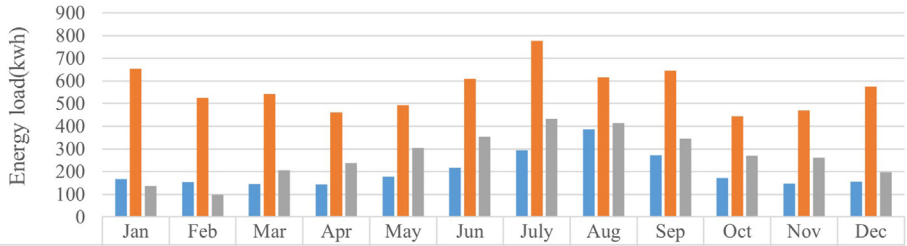


Fig. 18. The indoor air temperature and the percentage of comfort time in different rooms of the case study.



	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec
Actual bill 1	167	155	146	144	178	216	293	387	272	171	147	156
Actual bill 2	654	525	543	462	493	610	777	615	646	445	469	574
Energy(Kwh)	137.88	99.56	206.42	237.32	305.11	353.05	432.65	414.81	344.84	269.47	261.46	197.23

Fig. 19. The comparison of the simulated energy consumption and energy bills of two households in the case study building (acquired from the field survey).

surveys with building energy modelling. Future works can also integrate smart meter data with models for a more detailed building energy prediction.

4. Conclusions and future works

As a case study, this research looked at a typical high-rise residential construction in southeast China, an example of a high-rise skyscraper built during China's rapid urbanization. The results are based on the foregoing analysis, emphasising the impact of occupancy behaviours on energy loads and CO_2 emissions. The connection between occupancy behaviour and energy usage is researched to identify the elements that influence it. Finally, a simulation of the occupancy behaviour profile is run to determine the most important issue influencing energy usage in the household.

The internal gains profiles and window-opening profiles were created using the questionnaire data and correlation analysis, and they reflect the lifestyle of people living in the high-rise dwellings in the target city. The statistical relationship between energy usage behaviour profiles and energy consumption in urban dwellings were studied, and simulations based on this data were run. The modelling of typical houses revealed that while occupancy behaviour was smaller than equipment, it was still important to improve.

With China's new “more than one child” law, family structures and sizes are likely to grow larger than previously, resulting in a significant rise in energy loads and CO_2 emissions for the new family compared to the household presented. Given the urbanization and population growth in Chinese cities, moving to an urban location will be a trend in the future. It will also improve commuting energy use. According to the results of the models, energy usage will rise per family in the future. The future development of a better living environment, improving energy efficiency or using renewable energy will be required to maintain energy usage under control. Different operational behaviours generate differences in energy usage, and authorities' designers should be aware of this trend.

For the future development of the residential sector, the diffusion of smart energy management systems would potentially improve energy efficiency. Combining heat pump, photovoltaic, battery and thermal storage (hot water tank and building mass), a household can optimize the operation of technologies for lower peak loads, less grid-electricity consumption, as well as lower energy cost [53].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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